

Tracking Energy Transfers and Transformations

PhET Energy Forms and Changes · PhET

Years 8-9

~30 minutes

Science · Physics

NOTICE — AI-generated worksheet

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Simulation: <https://phet.colorado.edu/en/simulations/energy-forms-and-changes>

Curriculum links: AC9S8U05 · AC9S9U05 · energy transfer

Learning intentions

- I can identify examples of kinetic, potential, thermal and electrical energy in a system.
- I can trace how energy transfers and transforms as it moves between objects.
- I can apply the law of conservation of energy to explain the efficiency of a simple energy system.

Getting started

Open the simulation and start on the Intro screen. Select an object such as the iron or the brick and place it on the heater or the ice to see energy symbols appear and flow. Once you've explored that, move to the Energy Systems screen for the later tasks.

Tasks

1. Place the iron on the heater. Describe what happens to the energy symbols as the iron heats up.

2. Now place the same iron on the ice block instead. What happens to the flow of energy symbols this time?

3. Based on tasks 1 and 2, explain in your own words which way thermal energy always transfers between two objects at different temperatures.

4. Switch to the Energy Systems screen. Turn on the sun/solar panel chain. List, in order, each form of energy you can see as it moves from the sun to the final energy user (e.g. a light globe).

5. Try three different energy source and energy user combinations on the Energy Systems screen. For each, note whether the chain looks 100% efficient or whether some energy is visibly lost or wasted.

Energy source chosen	Energy user chosen	Fully efficient, or energy lost?

6. Using the law of conservation of energy, explain what happens to any energy that is 'lost' along a chain in your table — is it destroyed, or has it transformed into a less useful form?

7. Compare two different energy sources on the Energy Systems screen. Which one delivers energy to the user more directly, with fewer transformation steps?

Extension (optional)
Choose one everyday appliance (e.g. a kettle or a toaster) and describe its full energy transformation chain from the original energy source through to the useful energy output, including where some energy is likely wasted.
